CT32 | Exact Frobenius mod-49 certificate

Independent arithmetic replay, conditional on CT30-31 forward kernel identification

E109 (109 rational points)

trace t = -7; Frobenius roots mod 7: 2, 5
Unique lifts mod 49: 9, 33; chosen forward = 33
ord(33) = 42; smallest d with 33^d = +/-1 mod 49 is 21

E113 (113 rational points)

trace t = -11; Frobenius roots mod 7: 4, 6
Unique lifts mod 49: 11, 27; chosen forward = 27
ord(27) = 14; smallest d with 27^d = +/-1 mod 49 is 7

Combined two-hop geometric kernel

Full 49 x 49 torsion points need the extension F_101^42.
Coordinate-wise x-supports need only F_101^21.

Proof sketch

Frobenius on the 7-adic Tate module has polynomial X^2 - tX + 101.
Distinct roots mod 7 lift uniquely to mod 49 by Hensel. For a generator Q
of the selected cyclic order-49 kernel, Frobenius(Q) = lambda Q.
The full point field requires lambda^d = 1; x(Q) is fixed when lambda^d = +/-1.
Direct enumeration independently checks the exact multiplicative orders.
Only 7 of the 49 kernel points of either factor are fixed over F_101^6.
The combined point kernel has 2401 elements; just 49 are fixed over F_101^6.

Scope and evidence

Producer-local Python and independent Go arithmetic executed; negative controls passed.
No signed independent PCC verdict; no Jacobian map or solving-cost speedup here.
The F_101^21 x-only representation is an unbenchmarked implementation option.
Root: isogeny-labs/crypto PR #3877 | Supporting oracle: cryptanalysis PR #1492
Established theory: Kohel (1996); Sutherland 18.783 (2025).